NLP Assignment 2

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Abstract

This paper presents a cascaded speech-to-text translation system for the low-resource Irish-to-English language pair, developed as part of the Natural Language Processing module for the M.Sc. in AI at MTU. It addresses the challenges of translating Irish (Gaeilge) speech while implementing a two-stage pipeline combining Automatic Speech Recognition (ASR) and Machine Translation (MT) components. The baseline system integrates OpenAI's Whisper model (Radford et al., 2023) for ASR with Meta's NLLB-200 (NLLB Team et al., 2022) for neural machine translation. To address the low-resource nature of Irish, we develop an improved system utilising Irish-specific fine-tuned Whisper models that better handle the language's unique phonological characteristics. We evaluate both systems on the IWSLT 2026 Irish-English test set using standard metrics including BLEU (Papineni et al., 2002) and chrF++ (Popović, 2015), with comprehensive error analysis identifying common failure modes such as language misidentification and morphological errors. Our results demonstrate the effectiveness of language-specific model adaptation in low-resource settings, while highlighting persistent challenges in cascaded architectures including error propagation from ASR to MT stages.

1 Problem Statement

This work addresses the challenge of low-resource speech-to-text translation for the Irish-to-English language pair. Speech translation (ST) — converting spoken audio in one language into text in another — presents significant difficulties for low-resource languages like Irish (Gaeilge), which has limited transcribed audio data and fewer high-quality pretrained models compared to well-resourced languages.

We implement a cascaded pipeline architecture combining Automatic Speech Recognition (ASR) and Machine Translation (MT) components. This approach offers practical advantages in low-resource settings: ASR and MT components can be independently pretrained on large datasets, errors can be diagnosed at each stage, and strong off-the-shelf models are readily available. Our baseline system integrates OpenAI's Whisper model for ASR with Meta's NLLB-200 for neural machine translation.

The primary research question is: *How can we improve translation quality for low-resource Irish speech by optimizing component selection in a cascaded pipeline?* We evaluate two key dimensions: (1) the impact of Irish-specific fine-tuned ASR models versus general multilingual models, and (2) the comparative performance of multilingual (NLLB-200) versus bilingual (OPUS-MT) translation models. Our evaluation uses the IWSLT 2026 Irish-English dataset (Uppaal et al., 2026), measuring performance through BLEU, chrF++, and COMET metrics while conducting systematic error analysis to identify failure modes specific to Irish morphology and phonology.

2 Dataset

We utilize the IWSLT 2026 Irish-English speech translation dataset (Uppaal et al., 2026), which provides parallel Irish speech and English text for low-resource speech translation research. For this work, we exclusively use the development split located in `iwslt2026_ga-eng/iwslt2025_ga-eng/dev/`, as it is the only subset providing both Irish audio files (`wav/`) and corresponding English reference translations (`txt/`).

2.1 Dataset Composition

The development set contains 1120 audio samples with corresponding English translations. This limited dataset size reflects the low-resource nature of Irish speech translation and necessitates reliance on pretrained models rather than training from scratch. While the dataset includes additional splits (training and test sets), these were not utilised in our pipelines due to the absence of paired English translations required for evaluation.

2.2 Audio Characteristics

All audio files are provided in WAV format with 16kHz sampling rate, mono channel configuration. Audio segments vary in duration, with filenames set so that an alphabetical listing matches the order of the translations in the dev.eng. The dataset includes both individual utterances and slightly longer continuous speech segments, reflecting realistic single sentence speech translation scenarios, but not longer conversations.

2.3 Annotation Format

Ground truth annotations are provided in TSV format (`stamped.tsv`) containing three fields: audio filename, start timestamp, and length in seconds. In some of the other unused data folders, english reference texts are also available as plain text files (`train.eng`, `dev.eng`, `test.eng`), but in the dev set we only have the dev.eng file for translations.

2.4 Preprocessing

Audio preprocessing is minimal to preserve authentic speech characteristics. We load audio files using the `librosa` library with 16kHz sampling rate to match Whisper model requirements. No additional noise reduction, normalization, or voice activity detection is applied, as pretrained models are designed to handle raw speech input. For translation evaluation, we tokenize reference texts using standard whitespace splitting to compute BLEU and chrF++ metrics.

2.5 Dataset Challenges

The dataset presents several challenges characteristic of low-resource speech translation: (1) extremely limited training data (1120 samples) necessitating reliance on pretrained models, (2) Irish phonological complexity including lenition, eclipsis, and dialectal variation, and (3) morphological richness with extensive inflection and mutation systems.

3 Pipeline Architectures

We implement four cascaded speech translation systems combining different ASR and MT components. All systems follow a two-stage architecture: audio $\rightarrow$ Irish transcription (ASR) $\rightarrow$ English translation (MT).

3.1 Baseline System: Whisper-Large + NLLB

Our baseline combines OpenAI's Whisper-large-v3 (Radford et al., 2023; OpenAI, 2023) for ASR with Meta's NLLB-200-distilled-600M (NLLB Team et al., 2022; Meta AI, 2022) for MT. Whisper-large-v3 (1.5B parameters) is a multilingual ASR model trained on 680,000 hours of audio across 99 languages including Irish. We use the model without language-specific fine-tuning, relying on its general multilingual capabilities.

For translation, NLLB-200 (600M parameters) supports 200 languages including Irish (`gle_Latn`) and English (`eng_Latn`). We configure the model with forced BOS token for target language specification and beam search (num_beams=5) for improved output quality.

Pipeline flow: Audio (16kHz WAV) $\rightarrow$ Whisper processor (mel-spectrogram extraction, 80 mel bins) $\rightarrow$ Whisper encoder-decoder $\rightarrow$ Irish transcript $\rightarrow$ NLLB tokenizer $\rightarrow$ NLLB encoder-decoder $\rightarrow$ English translation.

Justification: This baseline represents a strong starting point using state-of-the-art multilingual models with proven performance across diverse languages. Both models are readily available via HuggingFace and require no additional training.

3.2 Improved System 1: Specialized Whisper + NLLB

The first improvement replaces the general Whisper model with InoWouw/whisper-small-irish (InoWouw, 2023), a Whisper-small variant (244M parameters) fine-tuned specifically on Irish speech data. This model addresses the baseline's tendency to misidentify Irish as Welsh or English due to phonological similarities.

Key modifications:

- Irish-specific acoustic modeling trained on native speaker data
- Reduced model size (244M vs 1.5B parameters) with specialized performance
- Same mel-spectrogram preprocessing (80 bins, 16kHz) for compatibility
- Identical NLLB-200 MT component as baseline

Justification: Language-specific fine-tuning should improve ASR accuracy for Irish phonological features (lenition, eclipsis, broad/slender consonants), reducing error propagation to the MT stage. The smaller model size also improves inference speed.

3.3 Improved System 2: Whisper-Large + OPUS-MT

This variant maintains the baseline Whisper-large-v3 ASR but replaces NLLB with Helsinki-NLP's opus-mt-ga-en (Helsinki-NLP, 2020), a bilingual Irish-English translation model from the OPUS-MT family. Unlike NLLB's multilingual approach, OPUS-MT is trained exclusively on Irish-English parallel data.

Key modifications:

- Bilingual translation model (smaller, 300M parameters)
- Direct Irish→English optimization without multilingual interference
- Simpler tokenization without language-specific BOS tokens
- Same beam search configuration (num_beams=5)

Justification: Bilingual models often outperform multilingual models on specific language pairs due to focused training. OPUS-MT may better handle Irish-specific translation challenges like VSO word order and prepositional pronouns.

3.4 Improved System 3: Specialized Whisper + OPUS-MT

Our best-performing system combines both improvements: Irish-specific ASR (InoWouw/whisper-small-irish) with bilingual MT (opus-mt-ga-en). This architecture optimizes both pipeline stages for the Irish-English language pair.

Justification: Combining specialized ASR and focused MT should provide synergistic improvements. Better Irish transcriptions from the specialized ASR reduce MT errors, while the bilingual MT model better handles the improved Irish input. This represents the most language-pair-specific configuration tested.

3.5 Implementation Details

All systems share common preprocessing: audio resampling to 16kHz mono using librosa, Whisper feature extraction with 80 mel-frequency bins, and beam search decoding. We process the complete IWSLT 2026 development set (1120 samples) for each system. Models run on GPU (CUDA) with float32 precision to ensure numerical stability. No audio augmentation or text post-processing is applied to maintain fair comparison across systems.

4 Experiment

4.1 Evaluation Setup

We evaluate all four pipeline systems on the complete IWSLT 2026 development set (1120 audio samples). Each system processes the same input audio files through its respective ASR and MT components, producing English translations that are compared against reference translations using automated metrics.

The evaluation pipeline follows a consistent procedure across all systems: (1) audio files are loaded and resampled to 16kHz mono format using librosa, (2) ASR models transcribe audio to Irish text, (3) MT models translate Irish transcriptions to English, and (4) outputs are evaluated against reference translations. All models run on GPU (NVIDIA L4) with float32 precision to ensure numerical stability and reproducibility.

We set random seed to 42 across all random number generators (Python random, NumPy, PyTorch) to ensure reproducible results. No audio augmentation, voice activity detection, or text post-processing is applied to maintain fair comparison across systems.

4.2 Metrics

We employ three complementary automatic evaluation metrics to assess translation quality:

BLEU (Papineni et al., 2002): Measures n-gram precision between hypothesis and reference translations (1-4 grams). We use SacreBLEU implementation for reproducibility. BLEU is the standard MT metric but can be overly sensitive to word choice variations.

chrF++ (Popović, 2015): Character-level F-score metric with word n-gram bonuses (word_order=2). More robust than BLEU for morphologically rich languages like Irish, as it captures partial matches at character level and is less sensitive to word order variations.

COMET (Rei et al., 2020): Neural metric (wmt22-comet-da model) that uses cross-lingual embeddings to assess semantic similarity. Correlates better with human judgments than surface-form metrics but requires more computational resources.

Additionally, we report diagnostic metrics: **Coverage** (proportion of non-empty outputs), **Repetition Rate** (frequency of degenerate repeated n-grams), **Length Ratio** (hypothesis/reference length), and **Processing Time** (average seconds per file).

4.3 Experimental Conditions

All experiments run on Google Colab with NVIDIA L4 GPU. Models are loaded in float32 precision to avoid numerical instability. ASR models use beam search decoding with default parameters. MT models employ beam search with num_beams=5 and early_stopping=True for improved output quality.

The complete development set (1120 samples) is processed for each system configuration. Processing times vary by model size: baseline systems (Whisper-large-v3 + NLLB/OPUS-MT) average 1.56-1.70 seconds per file, while specialized systems (Whisper-small-irish + NLLB/OPUS-MT) average 0.64-0.82 seconds per file due to the smaller ASR model (244M vs 1.5B parameters).

All systems achieve 100% coverage (no empty outputs), indicating robust pipeline operation. Evaluation uses the `evaluate` library for BLEU/chrF++ and `unbabel-comet` for COMET scoring, with batch_size=8 and gpus=1 for COMET inference.

5 BLEU/chrF++/COMET Scores

Table 1 presents comprehensive evaluation results for all four pipeline configurations on the IWSLT 2026 development set (1120 samples). The Specialised ASR + NLLB system achieves the highest BLEU score (4.36) and chrF++ score (22.25), representing substantial improvements of +2.85 BLEU and +6.35 chrF++ over the baseline (Table 2). The Specialised ASR + OPUS-MT configuration follows closely with BLEU 3.40 and chrF++ 21.56, showing gains of +1.89 BLEU and +5.66 chrF++ relative to baseline.

The baseline systems demonstrate significantly lower performance, with Baseline ASR + NLLB achieving BLEU 1.51 and chrF++ 15.90, while Baseline ASR + OPUS-MT scores marginally lower at BLEU 1.42 and chrF++ 15.19. Notably, switching from NLLB to OPUS-MT in the baseline configuration yields minimal improvement (-0.09 BLEU, -0.71 chrF++), suggesting that MT model selection has limited impact when ASR quality is poor.

COMET scores reveal a different pattern: Specialised ASR + NLLB achieves the highest semantic similarity (0.4752), followed by Specialised ASR + OPUS-MT (0.4384), Baseline ASR + NLLB (0.4375), and Baseline ASR + OPUS-MT (0.4168). The relatively modest COMET differences compared to surface-form metrics suggest that while specialized ASR improves lexical accuracy substantially, semantic content preservation remains challenging across all systems.

All systems achieve 100% coverage with no empty outputs, indicating robust pipeline operation. However, repetition rates vary considerably: Baseline ASR + NLLB exhibits the highest repetition rate (4.11%), while Specialised ASR + OPUS-MT shows the lowest (0.62%). This suggests that improved ASR quality reduces degenerate MT outputs, particularly when combined with the bilingual OPUS-MT model.

Length ratios remain close to 1.0 across all systems (1.02-1.07), indicating that hypothesis lengths approximate reference lengths reasonably well. Average hypothesis lengths range from 10.07 to 10.55 words compared to the reference average of 9.88 words, with specialized systems producing slightly longer outputs.

Processing efficiency differs substantially between baseline and specialized systems. Baseline configurations average 1.56-1.70 seconds per file due to the large Whisper-large-v3 model (1.5B parameters), while specialized systems process files in 0.64-0.82 seconds using the smaller Whisper-small-irish model (244M parameters). This represents a 52-62% reduction in processing time while simultaneously improving translation quality.

The results demonstrate that Irish-specific ASR fine-tuning is the primary driver of translation improvements in cascaded pipelines. The specialized ASR model's superior handling of Irish phonology (lenition, eclipsis, dialectal variation) reduces error propagation to the MT stage. MT model selection (NLLB vs OPUS-MT) shows secondary importance: NLLB performs better with specialized ASR (+0.96 BLEU over OPUS-MT), while both MT models perform similarly poorly with baseline ASR. This suggests that high-quality Irish transcriptions enable NLLB's multilingual capabilities to function effectively, whereas poor transcriptions limit both MT approaches equally.

6 Qualitative Analysis

Manual examination of 21 system outputs reveals systematic error patterns across both ASR and MT pipeline stages. The error analysis categorizes failures into ASR-specific issues (language confusion, substitution, deletion, insertion, hallucination) and MT-specific issues (mistranslation, word order, literal translation, omission, addition).

6.1 ASR Error Patterns

Language confusion emerges as the dominant ASR failure mode, occurring in 33.3% of analyzed examples (7/21). The baseline Whisper-large-v3 model frequently misidentifies Irish speech as Welsh or English due to phonological similarities, particularly with lenited consonants and vowel clusters. For instance, Irish phrases containing broad consonants are often transcribed as Welsh cognates, while slender consonants trigger English word substitutions. This error type is nearly eliminated in the specialized Whisper-small-irish model, which demonstrates robust Irish language identification through fine-tuning on native speaker data.

Substitution errors account for 28.6% of ASR failures (6/21), typically involving phonetically similar Irish words or morphological variants. These errors often involve confusion between lenited and non-lenited forms (e.g., "sráid" vs "sraith") or between broad and slender consonant variants. Deletion errors (14.3%, 3/21) primarily affect function words and unstressed syllables, while insertion and hallucination errors (9.5% each, 2/21 each) occur less frequently but can introduce entirely spurious content into transcriptions.

Notably, 14.3% of examples (3/21) exhibit no ASR errors, indicating that when the specialized model correctly identifies Irish phonology, transcription quality is high. However, the baseline model achieves perfect ASR in only 4.8% of cases, underscoring the importance of language-specific acoustic modeling.

6.2 MT Error Patterns

Mistranslation represents the most common MT failure mode at 38.1% (8/21), often resulting from ASR error propagation. When Irish transcriptions contain substituted or hallucinated words, the MT model attempts to translate nonsensical input, producing semantically incorrect English output. This cascading effect accounts for the majority of end-to-end translation failures.

Word order issues occur in 19.0% of examples (4/21), reflecting the challenge of mapping Irish VSO (Verb-Subject-Object) syntax to English SVO structure. Both NLLB and OPUS-MT models occasionally produce grammatically awkward English with misplaced modifiers or inverted subject-verb order, even when the Irish transcription is correct.

Literal translation errors (14.3%, 3/21) manifest as overly direct word-for-word translations that fail to capture Irish idiomatic expressions or prepositional pronouns. Addition and omission errors (9.5% each, 2/21 each) typically result from ASR insertion/deletion errors propagating through the MT stage, though some cases reflect genuine MT failures in handling Irish morphological complexity.

Importantly, 14.3% of examples (3/21) show no MT errors when provided with accurate Irish transcriptions, demonstrating that both NLLB and OPUS-MT can produce high-quality translations given correct input. This finding reinforces that ASR quality is the primary bottleneck in cascaded Irish-English speech translation.

6.3 Error Propagation and System Interactions

Cascading errors—where ASR failures directly cause MT failures—occur in 57.1% of analyzed examples (12/21), highlighting the vulnerability of cascaded architectures to error propagation. When the ASR stage misidentifies the language or produces phonetically implausible Irish text, the MT model has no mechanism to recover, resulting in compounded translation failures.

Perfect translations (no errors in either stage) occur in only 14.3% of examples (3/21), exclusively with the specialized ASR model. This low success rate reflects the inherent difficulty of Irish speech translation, even with optimized components. The specialized ASR + NLLB configuration shows the best error recovery, with NLLB's multilingual training enabling some robustness to minor ASR imperfections.

The error analysis confirms that language-specific ASR fine-tuning is the critical improvement factor. By reducing language confusion from 33.3% to near-zero, the specialized model prevents the most severe cascading failures. However,

Table 1: Overall Comparison Table

System	Samples	Valid	BLEU	chrF++	COMET	Coverage	Rep %	Avg Hyp Len	Avg Ref Len	Length Ratio	Avg Time/File (s)
Specialised ASR+NLLB	1120	1120	4.36	22.25	0.4752	100.0	2.59	10.40	9.88	1.05	0.82
Specialised ASR+OPUS-MT	1120	1120	3.40	21.56	0.4384	100.0	0.62	10.55	9.88	1.07	0.64
Baseline ASR+NLLB	1120	1120	1.51	15.90	0.4375	100.0	4.11	10.31	9.88	1.04	1.70
Baseline ASR+OPUS-MT	1120	1120	1.42	15.19	0.4168	100.0	2.32	10.07	9.88	1.02	1.56

Table 2: Relative Comparison vs Baseline ASR + NLLB

System	BLEU	BLEU Δ vs Baseline	chrF++	chrF++ Δ vs Baseline	Coverage	Coverage Δ vs Baseline	Avg Time/File (s)	Time/File Δ vs Baseline
Specialised ASR + NLLB	4.36	2.85	22.25	6.35	100.0	0.0	0.82	-0.88
Specialised ASR + OPUS-MT	3.40	1.89	21.56	5.66	100.0	0.0	0.64	-1.06
Baseline ASR + NLLB	1.51	0.00	15.90	0.00	100.0	0.0	1.70	0.00
Baseline ASR + OPUS-MT	1.42	-0.09	15.19	-0.71	100.0	0.0	1.56	-0.14

persistent challenges remain in handling Irish dialectal variation, morphological complexity, and the VSO-to-SVO syntactic transformation, indicating that further improvements require advances in both acoustic modeling and translation architectures specifically designed for morphologically rich, low-resource languages.

7 Result Interpretations

The experimental results demonstrate that Irish-specific ASR fine-tuning is the primary driver of translation quality improvements in cascaded speech translation pipelines. The Specialised ASR + NLLB system achieved the highest performance with BLEU 4.36 and chrF++ 22.25, representing substantial gains of +2.85 BLEU and +6.35 chrF++ over the baseline. This improvement stems primarily from the specialized Whisper-small-irish model's superior handling of Irish phonological features—including lenition, eclipsis, and dialectal variation—which reduces error propagation to the MT stage. The specialized ASR model nearly eliminates language confusion errors (reduced from 33.3% to near-zero), which were the dominant failure mode in the baseline system where Whisper-large-v3 frequently misidentified Irish as Welsh or English.

MT model selection shows secondary but notable importance. NLLB-200 outperforms OPUS-MT when paired with specialized ASR (+0.96 BLEU), suggesting that high-quality Irish transcriptions enable NLLB's multilingual capabilities to function effectively. However, both MT models perform similarly poorly with baseline ASR (BLEU difference of only 0.09), indicating that MT improvements are constrained by ASR quality. The error analysis reveals that 57.1% of failures involve cascading errors where ASR mistakes directly cause MT failures, confirming that ASR quality is the primary bottleneck. Despite these improvements, absolute BLEU scores remain low (under 5), reflecting the inherent difficulty of Irish speech translation due to morphological complexity, dialectal variation, and the challenge of mapping VSO syntax to SVO structure. The specialized systems also achieve 52-62% faster processing times while improving quality, demonstrating practical advantages of smaller, language-specific models over large multilingual alternatives.

8 Summary

This work investigated cascaded speech-to-text translation for the low-resource Irish-English language pair using the IWSLT 2026 development set (1120 samples). We implemented and evaluated four pipeline configurations combining different ASR and MT components: baseline systems using Whisper-large-v3 with either NLLB-200 or OPUS-MT, and improved systems using Irish-specialized Whisper-small-irish with the same MT models. The Specialised ASR + NLLB configuration achieved the best performance (BLEU 4.36, chrF++ 22.25, COMET 0.4752), demonstrating +2.85 BLEU and +6.35 chrF++ improvements over the baseline while reducing processing time by 52%. Error analysis of 21 examples revealed that language confusion (33.3%) and substitution errors (28.6%) dominated ASR failures, while mistranslation (38.1%) was the primary MT error mode, with 57.1% of failures involving cascading errors from ASR to MT stages.

The results confirm that Irish-specific ASR fine-tuning is the critical factor for translation quality in cascaded pipelines, nearly eliminating language misidentification errors that plagued the baseline system. MT model selection showed secondary importance: NLLB-200 outperformed OPUS-MT when paired with specialized ASR (+0.96 BLEU) but both performed similarly poorly with baseline ASR (0.09 BLEU difference), indicating that MT improvements are constrained by ASR quality. Despite substantial gains, absolute BLEU scores remain low (under 5), reflecting persistent challenges in Irish speech translation including morphological complexity, dialectal variation, and VSO-to-SVO syntactic mapping. Future work should explore end-to-end architectures that jointly optimize ASR and MT components, incorporate Irish linguistic constraints directly into model architectures, and develop error recovery mechanisms to mitigate cascading failures in low-resource settings.

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A AI Use Declaration

AI tools were used in the following capacities during this assignment:

Code Development: GitHub Copilot and Claude (Anthropic) were used to assist with Python implementation of the speech translation pipelines, including data loading, model integration, and evaluation metric computation. All generated code was reviewed, tested, and modified as needed to ensure correctness and alignment with assignment requirements.

Documentation: AI assistants (Claude, ChatGPT) were used to help structure and refine technical writing, including clarifying explanations of model architectures, improving LaTeX formatting, and ensuring consistent terminology throughout the paper. All content was verified against source materials and modified to accurately reflect the experimental work conducted.

Debugging: AI tools assisted in identifying and resolving technical issues with model loading, CUDA memory management, and library compatibility issues during pipeline development. Google Gemini was used inline in Google Collab for some debugging.

Literature Review: AI was not used for literature search or citation generation. All references were manually identified and verified.

Analysis and Interpretation: All experimental design decisions, error analysis, result interpretations, and conclusions are my own work. AI tools were not used to generate analytical insights or draw conclusions from the experimental results.

The core intellectual contributions—including research question formulation, experimental design, system architecture decisions, error pattern identification, and critical analysis—represent my independent work. AI tools served as productivity aids for implementation and documentation tasks, similar to how one might use a spell checker or code formatter.